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Farmers' traditional knowledge on climate change and weather forecast: The case of Menze Gera Midir district, Ethiopia

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ABSTRACT

Diverse sources of climate information are needed for smallholder farmers to help them reduce their vulnerability to adverse impacts of climate change. This study explored traditional knowledge (TK) of farmers on climate change and weather forecast in Menze Gera Midir district of Ethiopia. Data were collected from randomly selected 118 farmers. In addition, six focus group discussions and forty-five key informant interviews were held. Results indicated that farmers had intrinsic knowledge of forecasting weather and climate by using their lived experiences. Farmers' TK in terms of agricultural development was, however, limited. The study implied that the need for proper documentation of TK and its integration with scientific knowledge to address different environmental and socio-economic challenges of farming community.

1. Introduction

Traditional knowledge (TK) has been defined and redefined for a long period of time because of its growing significance in fields like geography (Smith, 2011), climate change and disaster management (Dekens, 2007), anthropology (Sillitoe, 1998), and developmental (Chambers et al., 2019). This body of knowledge is synonymously used as indigenous knowledge, and local knowledge (Joshi and Chelliah, 2013). The terms indigenous, traditional, and/or local knowledge make reference to knowledge that is accumulated over generations and guides societies in their numerous interactions with their immediate environment (Ajayi and Mafongoya, 2017).

Diverse sources of climate information are needed for effective and planned climate change adaptation to address vulnerable smallholder farmers (Zvobgo et al., 2023). In Ethiopia where weather stations are limited in number, unevenly distributed and located only in towns, farmers' TK is the major source of climate information for the vast rural areas of the country (Alemayehu and Bewket, 2017). Despite many smallholder farmers are using TK globally to forecast weather and climate, little is known about how the use of these forecasts is included in decisions and actions for mitigating climate risks (Ajayi and Mafongoya, 2017; Chambers et al., 2019; Zvobgo et al., 2023). Examining the role of TK in weather forecast and climate adaptation decision-making of smallholder farmers is essential (Ajayi and Mafongoya, 2017; Chambers et al., 2019; Mugi-Ngenga et al., 2021; Nyadzi et al., 2022; Zvobgo et al., 2023). There is a growing interest in the potential role of TK in adaptation to climate change (Newsham and Thomas, 2011; Zvobgo et al., 2023). Its systems are now recognized by several institutions, including the Intergovernmental Panel on Climate Change (IPCC), as an imperative participatory forecast approach for decision-making, particularly at the farm level (Chand et al., 2014; Ajayi and Mafongoya, 2017). TK plays an integral role in building climate resilience and is considered an alternative way of promoting development in poor rural communities in many parts of the world (Ayal et al., 2015).

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Farmers and pastoralists in Africa use combinations of biological, meteorological, and astrological indicators to predict seasonal weather conditions (Chambers et al., 2019; Radeny et al., 2019; Nyadzi et al., 2022; Zvobgo et al., 2023). Before the introduction of modern meteorological instruments, many traditional societies depended merely on TK systems to forecast weather through several years of observation (Rautela and Karki, 2015).

However, recent studies have grown increasingly pessimistic about the viability of traditional weather forecasting using the TK of farmers (Risiro et al., 2012; Makwara, 2020). Some of the concerns included the declining popularity of traditional weather forecasting (Roncoli et al., 2002; Kipkorir et al., 2010), expansion of modern education and monotheistic religions (Risiro et al., 2012; Guye et al., 2022), growing generation gap and (Guye et al., 2022), lack of clear TK transfer mechanisms (Ifejika Speranza et al., 2010; Makwara, 2020), and lack of coordinated research on accuracy and reliability of TK (Radeny et al., 2019). Moreover, scientific knowledge has increasingly taken priority over local knowledge and practice in agricultural systems research and development in recent decades (Ajayi and Mafongoya, 2017). Modern science has not come up with a conclusive stance for or against the claims of traditional weather forecasting (Mundy and Compton, 1991). Incorporation of traditional forecast methods into modern forecast systems significantly improves the communication and application of climate information easily (Plotz et al., 2017; Mugi-Ngenga et al., 2021).

Ethiopia has unique culture, religion, customs, and traditions. There are tendencies of the scientific community to use and integrate TK into development planning. The recently launched homegrown economic reform and green legacy can be taken as examples. However, there are limited studies on the TK of farmers focusing on weather and climate issues. The available studies mainly focused on pastoralists in the arid and semi-arid areas of the country and did not comprehensively address all communities. For example, Ayal et al. (2015) studied the biotic weather forecast among the Borena herders of southern Ethiopia. Balehegn et al. (2019) focused on indigenous climate and weather forecast among Afar pastoralists. A study by Radeny et al. (2019) addressed the role of indigenous knowledge for seasonal weather and climate forecast in eastern Africa including Ethiopia. Iticha and Husen (2019) studied adaptation to climate change using indigenous weather forecast in Borana pastoralists. Likewise, Guye et al. (2022) investigated indigenous weather forecasting knowledge for drought monitoring among Gujii pastoralists in southern Ethiopia.

Climate variability is among the main threats to rainfed farming system in sub-Saharan Africa countries. Hence, knowledge on TK of farmers is important to design effective strategies for adaptation planning and vulnerability reduction (Mugi-Ngenga et al., 2021). TK by its nature differs from place to place, community to community, and from one agroecology to another (Mazzocchi, 2006; Mugi-Ngenga et al., 2021). TK is a non-scientific and non-quantifiable community-based knowledge based on observation of the behaviour of animals, plants, insects, wind patterns, movements and color of clouds, fillings and sensations on body parts, looking at

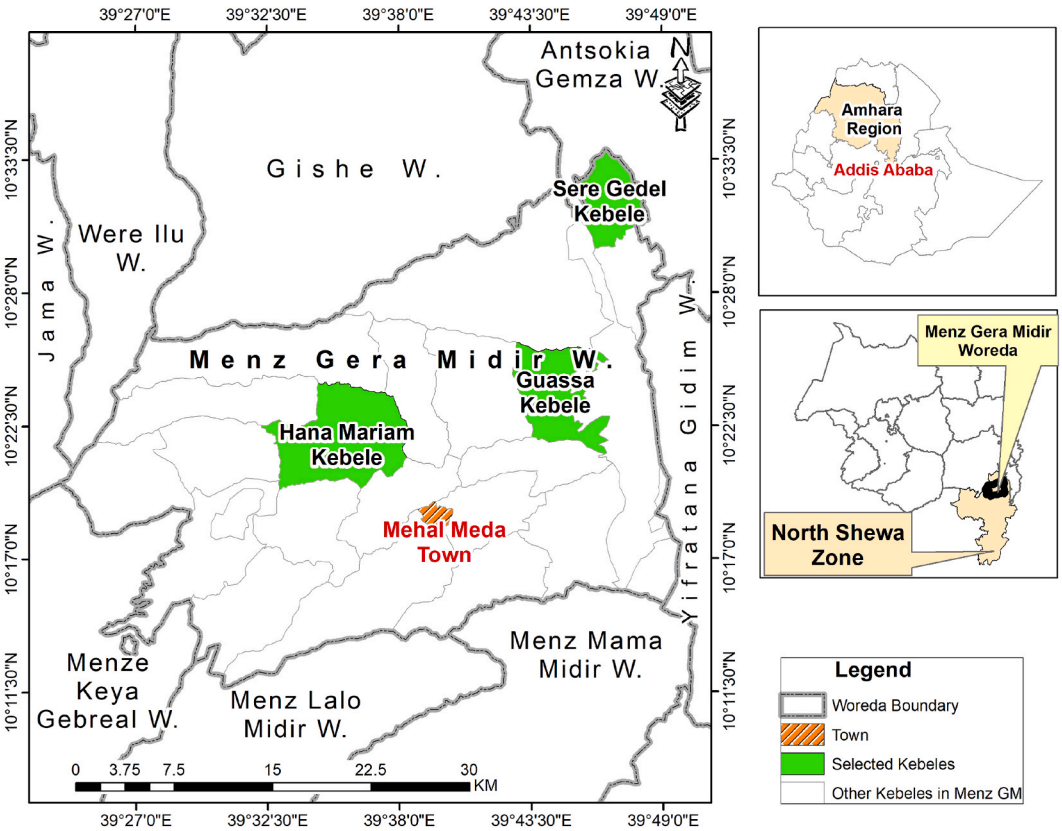

Fig. 1. Location of the study area.
Source: District shape file of CSA 2019 and ESRI imagery base map.

market conditions, lifestyles, interpreting alignment of sun, moon and stars, using religious calendars and other mechanisms. This study highlights the need to understand and document farmers' traditional knowledge of climate change and weather forecast for effective climate change adaptation and agricultural planning. Therefore, the objective of this study was to explore TK of farmers' on climate change and weather forecast in Menze Gera Midir district of Ethiopia. In the following section, we present a brief description of materials and methods, and this is followed by the results and discussion section. Conclusions are presented in Section 4.

2. Materials and methods

2.1. Description of the study area

The study was conducted in Menze Gera Midir district, central highland of Ethiopia (Fig. 1). The climate of the area is driven by the moisture from the Indian Ocean, Equatorial East Pacific, Gulf of Guinea, Mediterranean region, and Arabian Peninsula (Viste and Sorteberg, 2013). It has an annual rainfall ranging between 800 and 1390 mm; while the annual maximum and minimum temperature range between 9 °C and 24 °C, respectively (Worku et al., 2019). Mixed farming is the dominant livelihood in the area. Menze Gera Midir district belongs to one of the most vulnerable areas to climate change and variability in the country; where a large segment of its population is already food insecure (Alemayehu and Bewket, 2016).

2.2. Data collection and analysis

Data used for this study were collected from three different Kebeles (the lowest administration tire in the country) with diverse agroecological systems. Guassa is situated in the High Dega (3200–3700 m asl) agro-ecological zone, while Hana-Maryam and Siregedel are found in the Dega (2300–3200 m asl) and Woyna Dega (1500–2300 m asl) zones, respectively (Hurni et al., 2016).

Data were obtained from household surveys, key informant interviews (KIIs) and focus group discussions (FGDs) in the three Kebeles. In the first stage, Menze Gera Midir district and three representative samples Kebeles incorporating High Dega, Dega, and Woyna Dega agroecological zones were selected purposively to capture variations across agroecology. Frequent exposure to climate extremes, previous experience on the availability of traditional weather forecasting, and unique culture of arts, environmental conservation, cloth making, and traditional medicine were considered. The district is also food insecure, productivity is subsistence, and land degradation and declining soil fertility contributed to poverty. In the second stage, sample households were selected using a simple random sampling technique which gives equal chance to be selected for all households. In total, 118 households were selected randomly (33 in Guassa, 45 in Hana-Maryam, and 40 in Siregedel).

Participants of the FGDs and KIIs were purposively selected through agricultural extension officers and Kebele leaders based on their knowledge of TK. Accordingly, data on signs and observations of weather systems, weather forecast, indicators of extreme events, interpretation and precision of TK, community perception on TK weather forecast, and use of forecast information for agricultural decisions were collected. Forty five KIIs having fifteen participants per Kebele were used from priests, community leaders, agricultural extension workers, community based organizations like *Idir* (a traditional charity institution) leaders. Two FGDs having 10 participants per Kebele were conducted. FGD participants included elderly farmers with rich farm experience and TK. Following Mihas (2023), we used thematic analysis to analyse data obtained from qualitative methods. It is preferred over the other techniques when there is large body of data obtained from key informant interviews and focus group discussions. Data obtained from the household surveys were analysed using descriptive statistics.

3. Results and discussion

3.1. Background of respondents

The average age of the respondents is 60.0 years; indicating rich experience in TK related to climate change and weather forecast. Educational levels are low; more than 88% of households attended below secondary school educational level (Table 1). TK of climate

Table 1
Background information of respondents (% of respondents).

Item	Guassa	Hana-Maryam	Siregedel
Average age (in years)	62	58	60
Education level			
Can read and write	36	40	45
Cannot read and write	39	33	33
Primary (Grade 1–8)	18	20	15
Secondary (Grade 9–12)	6	7	8
Source of income			
Mixed farming	9	64	15
Cash crops	6	13	35
Trees	40	7	33
Off-farm activities	46	16	18

and weather forecast may not be necessarily related to modern educational level of farmers. This is because in TK, farmers gained lots of practical knowledge through trial and error accumulated for hundreds and thousands of years' experience. Educated farmers enhance their ability to receive, interpret and comprehend information relevant to making innovative decisions in their farms (Ndambiri et al., 2013). Studies in different African countries including Ethiopia confirmed that educated farmers have better perception and adaptation potential to climate change (Deressa et al., 2009; Alemayehu and Bewket, 2017).

In Guassa off-farm activities are farmers' main income source (45.5%). Whereas in Hana-Maryam and Siregedel mixed farming (64.4%) and cash crops (35%) are the main sources of income, respectively. All the respondents have their own farmlands. However, the average farm size is about 1.0 ha, which is below the national average. In addition, farmlands in many areas are fragmented and affected by soil erosion and declining soil fertility. Hence, it resulted in food insecurity. Our findings corroborate with a study by Alemayehu and Bewket (2016) that found farmland shortage and land fragmentation are major problems in the study area.

3.2. Farmers' source of weather and climate information

Farmers access weather and climate information from different sources such as media like radio, farmer-to-farmer extension, accessing extension service, and indigenous sources like observing the behaviour of animals, plants, winds and clouds, and other events. About 83% and 91% of the respondents in Guassa and Hana-Maryam respectively observe the behaviour of animals for the near future weather information. Radio (42% of respondents) and farmer-to-farmer extension (47% of the respondents) contributed as source of information in Hana-Maryam (Table 2). Observing plant behaviour is used by 46% of the respondents in Guassa. Participants in the FGDs and KIIs also reported that they come to the elders for asking information on what seems the near future weather conditions. The findings are similar to a previous study by Radeny et al. (2019) in the Borena area, southern Ethiopia where information from elders are useful.

3.3. Farmers' knowledge of climate change

In Guassa wildlife reserve, which is home to many endemic animals like, the Ethiopian wolf (*Canis simensis*), gelada baboon (*Theropithecus gelada*), and other animals like African wolves (*Canis aureus lupaster),* spotted hyenas (*Crocuta crocuta*), leopards (*Panthera pardus*), and Servals (*Leptailurus serval*) moved upward to optimum elevation in the eastern edge of the reserve which is an indication of climate change in the area. Following this, farmers anticipate changes in the climate condition particularly drought. Wildlife migration as an indicator of climate change is confirmed by 87% of the respondents in Hana-Maryam. More than 59%, of the respondents in Guassa and Siregedel also noted wildlife migration as an indicator of climate change. Our findings corroborate the findings of Guo et al. (2018) who reported that many species in different parts of the world shifted their geographic ranges toward higher elevations in response to rapid changes in temperature and precipitation regimes.

Respondents from Guassa reported that their area was foggy and they feel cold in all seasons; which now is clear or visibility except June–September (*Kiremt* season). Change in visibility of the environment an indicator of changes in climate has been confirmed by 91% of the respondents Guassa (Table 3). Moreover, more than half of the respondents in Siregedel confirmed that clouds during *Kiremt* season were thick and dense which now become thin and light which they consider that the climate is changing.

Respondents also replied that changes in the clothing style over time are an indicator of climate change. For example, three decades ago, the community wore very thick and furry clothes like *Gabi* (a cotton made cloth), *Bana* (a locally made blanket from of sheep fur), and blankets to protect themselves from the cold weather conditions especially from mid-July to mid-January. Currently, *Bana is* abandoned from the area and replaced by thin and short clothes which can be worn in all seasons. This indicates that the temperature of the area gets warming. In this regard, 83% of the respondents in Guassa understand climate change by changes in the clothing style. While only 25% of the respondents in Siregedel revealed that the clothing style in their area is changed due to climate change (Table 3).

Almost all respondents replied that the temperature of their local area is increasing from time to time which they associate with evidence of climate change. More than 70% of the respondents agreed that declining trends in annual and seasonal rainfalls and changes in the onset and offset periods an indicator of climate change (Table 3). Results from FGD and KIIs participants also explained a decline and absence of March–April (*Belg*) season rainfall and year to year variability of annual rainfall as indicators of climate change. Our findings corroborate the results of a previous study by Alemayehu and Bewket (2017).

About 82% and 91%of the respondents in Guassa and Hana-Maryam reported changes in the onset and cessation of rainfall period as an important indicator of climate change in their area. *Kiremt* rainfall onset period was around mid- June while the cessation was mid-September which now the onset and cessation periods are mid-July and end of August. This affects the cropping calendar. Similarly, FGD participants explained that there are changes in the annual and seasonal rainfall patterns which have implications for

Table 2
Farmers' source of weather and climate information (% of respondents).

Source of information	Guassa	Hana-Maryam	Siregedel
Radio	40	42	28
Accessing extension service	43	36	35
Observing animal behaviour	83	91	70
Observing plant behaviour	46	11	25
Farmer-to- farmers extension	60	47	58

Table 3
Indicators of climate change (% of respondents).

Indicators	Guassa	Hana-Maryam	Siregedel
Change in temperature	77	69	30
Change in dressing style	86	78	25
Wildlife migration	59	87	60
Extinction of plant species	71	18	68
Change in visibility	91	9	50
Decline in rainfall amount	76	84	70
Changes in onset and cessation of rainfall period	82	91	76

household level food security.

3.4. Farmers' traditional knowledge in weather forecasting

3.4.1. Direction and strength of winds

About 98% of the farmers in Guassa use wind direction for seasonal weather forecast (Table 4). If there are prolonged winds from Ephratana Gidim Woreda (southeast direction) from March to mid of April, with full confidence farmers are ready to sow cereals such as barley, wheat, bean, and pea. Farmers call the southeast sourced wind during *Belg* "*Yifat wind*" which is the place where the wind is coming from. On the other hand, farmers in Siregedel foresee rainfall if winds blow from Wejed (southern and southeastern). Moreover, farmers in Hana-Maryam anticipate rainfall if winds blow from Negase-Amba (southern) and the southern half part of Kuria (southeastern). In agreement with our findings, Iticha and Husen (2019) and Radeny et al. (2019) in their study areas reported farmers anticipate the characteristics of the coming seasons by observing the nature of winds.

3.4.2. Swirl winds

About 83% and 84% of respondents in Guassa and Hana-Maryam, respectively observe the direction of moving swirl wind to know the rainfall conditions of the coming season (Table 4). Swirls coming from the southeast direction towards the northwest are an indicator of the onset of *Belg* rainfall. Farmers perceive that if swirl winds are rolling in a clockwise direction, there will be sufficient rainfall while the anti-clockwise rotation indicates little or no rainfall and farmers shift to barley and lentil which demands little rainfall in the area. Almost all farmers in Hana-Maryam reported that if there is a hasty change of wind direction and the appearance of swirl winds, there is a probability of heavy rainfall. Similar studies by Mapfumo et al. (2016) and Radeny et al. (2019) reported that strong swirl winds indicate imminent rainfall onsets which are indicators for heavy rain to fall in their respective study areas.

3.4.3. Color and movements of clouds

Observation of cloud color and movement is one of the major mechanisms by which farmers predict seasonal weather conditions. Accordingly, about 77% (Guassa) and 89% (Hana-Maryam) of the respondents replied that they use cloud color and movement for weather forecast (Table 4). Farmers expect rain will come soon if the color of the clouds is black and greyish. Results of the KIIs and FGD participants also explained that heavy and black clouds are strong signs of rainfall while prolonged white color clouds indicate drought to appear in the near future. Previous studies in Africa reported that black clouds are matured and rain-bearing, dark clouds signify approaching heavy rain, and white and light clouds indicate drought season in the near future (Iticha and Husen, 2019; Gbangou et al., 2021).

Table 4
Indicators used for weather forecast (% of respondents).

Indicators	Guassa	Hana-Maryam	Siregedel
Direction and strength of winds	86	99	80
Swirl winds	83	84	73
Color and movements of clouds	77	89	68
Temperature change	74	73	75
Sound of hyena	94	78	35
Behaviour of birds	60	67	65
Behaviour of donkeys	69	51	58
Behaviour of chicken	77	62	35
Behaviour of cattle	91	93	73
Behaviour of ants	80	82.2	70
Behaviour of Butterflies	11	71	13
Behaviour of bees	54	69	88
Behaviour of plants	51	80	75
Changes observed in a human body condition	49	91	66
Observation at the coffee ceremony	83	67	78
Raphael market conditions	40	98	83

3.4.4. Temperature change

More than 73% of respondents use temperature variations (unusual hotness and coldness) to predict the coming weather conditions (Table 4). According to results from FGD participants, extreme warming towards the end of the dry season indicates a higher likelihood of a sufficient amount of rainfall. Hot temperatures are followed by high humidity and heavy rainfall conditions, cold temperatures accompanied by morning fog are considered as signs of bad weather conditions that will cause drought, calm weather with restricted wind movement and unusual high temperature indicates the onset of rain within the coming two weeks, and very high temperature at night is an indication of the likelihood of rainfall within 24 h. In line with our findings, Chengula and Nyambo (2016) and Makwara (2020) noted that temperature intensification during day and night is an alarm for near-future rain will fall within hours up to a day or two.

3.4.5. Hyena

The varying tones of hyena are used by farmers to make seasonal weather forecasts ranges from 35% of respondents (Siregedel) to 94% of respondents (Guassa) (Table 4). FGD and KII participants explained that if hyenas are screaming repeatedly with a "*wooow*" (soft and long sound) when moving from rivers and valleys to farmlands, it is an indication of heavy rainfall in the next two to four weeks. However, if the sound of hyena is "*hooom*" (with a babbling sound), it is a sign of drought in which there will not be sufficient rainfall in the coming weeks, particularly during the *Belg* season.

3.4.6. Birds

More than 60% of respondents use the behaviour of birds for weather forecast (Table 4). The FGD participants explained that if birds are singing by making first a circular or cone shape and then flying from one edge of the shape to another shape they made, it is an indication of a rainy and moist season. When birds are fighting each other, it is an indication of heavy rainfall with lightning and thunderstorms. When birds form a circular shape and singing when flying and repeatedly move vertically nearly touching the ground and towards the sky, it shows the rain will be heavy and sufficient for supporting growth of crops. Results of KIIs also noted that frequent congregation and singing of birds within a week, they perceive good rainy season will come soon. In addition, if birds are sitting on trees and singing during rainy hours, it is believed as the rainfall season would be prolonged and long-lasting. Whereas if birds are singing after rain when it is sunny, mostly it is considered that rain will stop recently. Similarity between the scientific forecasts and the bird phenology as indicator of the onset and cessation of rainfall was reported by Muguti and Maposa (2012), Chengula and Nyambo (2016), Radeny et al. (2019) and Nkuba et al. (2020).

3.4.7. Chicken

The use of chicken as an indicator of future weather condition ranges from 35% of the respondents (Siregedel) to more than 77% of the respondents (Guassa). Results from FGD and KII participants explained that when chickens come late to encamp in the evening; it is an indication of the likelihood of drought followed by hunger. When chickens come to encamp early before evening, good weather conditions and sufficient rain are expected. In addition, if a rooster frequently flutters its wings and screams like other bird species, rainfall is anticipated. If chickens spend nights in encamp with their wings outstretched and tickle their body throughout the night, and have a "*ku-ku-ku*" (with an unusual soft tone) repeatedly (for three or more days within a week), it is a sign of good rain. If chickens move far away from their homes to the farm in search of grain, it will be a sign of future drought. Participants of the FGD and KII also explained that the delay of chicken to go to encamp is not attributed to the availability of food. A study by Muguti and Maposa (2012) noted that if chickens move around when it is raining, it is an indication of plentiful rains in that particular season.

3.4.8. Cattle

The use of cattle to foretell the coming season's humidity, rainfall and drought conditions ranges from 76% of the respondents (Siregedel) to more than 94% of the respondents (Hana-Maryam). If oxen unusually are gamboling in the field, erect and flap their ears repeatedly by looking up to the sky, blowing and leaping to dry riverbeds and fighting each other, it is considered as a sign of good rainy season. On the other hand, if cattle scramble or fight for fodder, and eat what they are not usually eating like woody plant leaves; it is perceived that there will be drought in the coming season. On the other hand, if cattle lose appetite to eat fodder which was commonly consumed during drought periods, it indicates a good season to come soon. In this regard, our finding is contrary to the finding of Radeny et al. (2019) which reported a loss of appetite for grass and salt, are predictive for the coming drought season rather than good seasons among herders in east Africa including southern Ethiopia.

3.4.9. Donkeys

Donkeys are known to be weather predictors in the study area by more than half of the respondents. If donkeys are running by flapping their ears just to the leveled front, farmers normally anticipate rainfall. Results of the KIIs and FGDs reported that if donkeys bray by looking upward though there are no clouds on the sky, it will rain within one to two weeks. A similar study in east Africa showed that donkeys flapping their ears repeatedly at any time during the dry season indicate a near onset of rains, and prospects of a good season to come (Radeny et al., 2019).

3.4.10. Ants

Ants are perceived as predictors of occasions in general and of forthcoming weather conditions in particular by more than 70% of the respondents. Results from KIIs and FGDs discoursed that grey flying ants (*Odentotermes* spp) which are locally called *Kiremt-agbi* or rain bearer ants are true indicators of rainfall onset periods and occurrence of heavy rains. If a group of *Kiremt-agbi* are mounting from

the wet subsurface and infest the environment, *Kiremt* rainfall will onset most probably within a week. As a result, these ants are recognized as *News-Makers* by the community. If ants are collecting different grains and plant remains by making long lines and if the lines are non-interrupting, farmers know that there is long-staying rainfall. On the other hand, if ants are confused and deforming the lines and moving here and there, there will be unstable weather conditions; mainly strong winds that will dry up the land after rainy weeks. Studies showed that ants are used for seasonal weather forecast (Iticha and Husen, 2019; Radeny et al., 2019; Gbangou et al., 2021; Nkuba et al., 2020; Camacho-Villa et al., 2021).

3.4.11. Changes in seasons

Besides the rainy seasons, there are four seasons based on the Ethiopian Orthodox Tewahedo Church (EOTC) teachings nominated by the four evangelists (Matthew, Mark, Luke, and John) as years of Mathew, Mark, Luke, and John. When we divide the age of the earth to four and if the reminder is 0, the year is John; if the remainders are 1, 2, and 3, the years are Mathew, Mark, and Luke respectively. Here is an example that put the case in point. As per the EOTC teachings, Jesus Christ comes to this world after 5500 years after the world is created. After Jesus was born, it is 2015 years according to the Ethiopian calendar. Adding up the two we get 7515. While dividing this number by four, the remainder is 3, and hence the year is called "*Year of Luke*". Similarly, the years 2016, 2017, and 2018 in the Ethiopian calendar are the years of John, Mathew, and Mark respectively. Farmers have an inherent proverb called "*belukas eres beyohannes efes*" which means "plow (prepare land) in the year of Luke and harvest a huge amount of yield in the year of John". During the year of Luke, farmers are not confident to sow/plant crops; instead, they simply plow and make it ready for the next *Kiremt*.

3.4.12. Butterflies

Approximately 76% (Guass), 82% (Hana-Maryam) and 65% (Siregedel) of the respondents agreed on the use of butterflies' behaviour for weather forecast. Results from FGDs also revealed that the appearing and flying out of butterflies are indicators of the onset of drought conditions. If butterflies appear early in August, farmers will not remove weeds from their plots to prevent further drying of their plots as they did not anticipate rainfall during September which is useful for growing wheat. Our finding revealed that the flock appearance of butterflies after mid-August signifies the early cessation of *Kiremt* rainfall. Similar conclusions are reported on the use of butterflies as indicator of cessation of rain in different areas (Ifejika Speranza et al., 2010; Nkuba et al., 2020).

3.4.13. Bees

The behaviors of bees for future season weather forecast are used by 54% (Guassa) to 88% (Siregedel) of the respondents. By detecting the future drought, bees prepare more beeswax and honey to pass the hardship time by consuming what they collected and prepared. Accordingly, participants of FGDs explained that if the number of bees moving out from the beehive for collecting nectars suddenly increases in a dry season, a drought period in which dry up of flowering plants will occur. Previous studies by Galacgac and Balisacan (2009), Chang'a et al. (2010) and Speranza et al. (2010) noted bees help communities to in weather forecast.

3.4.14. Behaviour of plants

Flowering and fruiting of certain plants in unusual times is used as a signboard of the coming weather condition by about 80% (Hana-Maryam) and 75% (Siregedel) of the respondents (Table 4). *Amgne* (a local wheat variety that grows on vertisols during September) which is locally called "to be trusted on God" does not bear flowers every year during the dry season (December to January). However, when it bears flowers during the dry season, farmers trust the flowering of *Amgne* as an indicator of good rainfall conditions for *Belg*. Focus group discussants also noted that better rainfall is expected in the coming season after the flowering of *Amgne*. Similarly, when *Sama (Urtica dioica* L.*)* bears abundant fruits almost everywhere unusually in a dry season, it is an indication of convenient weather condition for beans and peas during Meher (the main cropping season). Use of flowering pattern of different plants is also used as an indicator of future climate in southern Ethiopia (Ayal et al., 2015) and Kenya (Ifejika Speranza et al., 2010).

3.4.15. Observed changes in the human body

Changes observed in the human body like palpitation on top of the right eye and one-sided headaches are used by more than 91% (Hana-Maryam) and 63% (Siregedel) of the respondents for weather forecast (Table 4). If a person itches his or her feet, he or she says, "*I will* stride on *wet*" or "I am to walk on the wet ground" which means it will rain in a near future. Different studies also affirm changes observed in the human body are used to forecast weather (Mahoo et al., 2015; Radeny et al., 2019).

3.4.16. Observation during a coffee ceremony

In Ethiopian culture, the coffee ceremony gathers neighbors all together where attendants discuss different issues including environmental issues and socio-economic issues. The flow pattern of boiled coffee helps to determine whether the coming rainfall is sufficient to support crop growth or not. If the coffee flows out of the coffee pot several times (often more than 3 times in a short duration), it indicates the onset of rain soon. Accordingly, observing what is happening in the coffee pot helps to anticipate the likelihood of getting rainfall in the near future by 83 (Guassa) and 67% (Hana-Maryam) of the respondents (Table 4).

3.4.17. Observing market conditions

There are 13 months in Ethiopia, and the 13th month (having 5 days in three consecutive years and 6 days in the fourth year i.e., *Year of John*) is called Pagumen, which means transitional. On the 3rd day of Pagumen, the annual feast of the Archangel St. Raphael is celebrated. The community gathered at the church called *Wuhey Ruphael* which is found in Mesele-Maryam Kebele of the district. The

wonderful thing is that it is not only to celebrate and be blessed by the feast they gathered, but also to observe the market in exchange for animals, grains, clothes, and other commodities coming from different parts of the district. If commodities and animals are sold at cheaper prices, they will remain cheap throughout the next year. For example, if oxen are sold at a low price, it is an indicator of bad weather conditions in the coming year where harvest may fail due to shortage of *Belg* rainfalls. Hence, farmers prepare their land for Meher season and reduce the number of cattle. Observing market conditions is largely used in Hana-Maryam (98% of the respondents) (Table 4). In addition, participants of KIIS and FGDs revealed that if rainfall occurs regardless of amount during the 3rd day of Pagumen which is the feast of Archangel St. Raphael and the Memorial Day of the second advent of Jesus Christ to this world, the coming year will be a blessing and a good harvest season. This rain is considered holy water that farmers will spray on their bodies, homes, barns, and farm grounds.

3.5. Challenges in using traditional knowledge

It is believed that farmers' TK on weather forecast fills gaps of scientific seasonal weather forecasts from meteorological observations to address the needs of farmers in planning and decisions making on resource utilization, conservation, and agricultural production. In spite of all the benefits, TK plays a limited role in addressing future climate. Hence, use and reliance by the local community faced certain challenges. First, the popularity and acceptance of TK declined over time as the young generation associates TK with conjuration (locally called *Tenqqola*). Second, the local government does not recognize TK as an important body of knowledge. Third, TK faced criticism from educated and civilized parts of the community on its reliability and validity as there is a lack of scientific evidence on the accuracy of TK. Fourth, the values of TK are not well documented and properly transferred to the younger generation. Fifth, TK lacks clear transfer due to poor documentation. Six, agricultural development workers and decision makers are not willing to hear and accept farmers' experiences. Even, advices are sometimes against the community expectations.

Regarding the reliability of TK, a 68-year-old man from Guassa, remembers about his village:

In our village, we mainly depend on TK for agricultural planning. We visited our fathers to discuss when to plow, saw and harvest and etc. Our fathers were perfect to forecast when the rain will start and end. That is how we survived for centuries in the absence of modern forecast. In addition, medical facilities were rare and we used traditional medicine mostly from plant leaves and roots from our surrounding. No one hesitated on the reliability of the medication. We are not perfect as did in our fathers. To the worst, our children are not only good in understanding future climate but also reluctant to use the information from us.

Regardless of these challenges, farmers of the study area use TK to mitigate the adverse impacts of climate change and variability using. A previous study by Alemayehu and Bewket (2017) on smallholder farmers' coping and adaptation strategies to climate change and variability in the central highlands of Ethiopia which is also the present study site identified different coping and adaptation strategies to climate and variability using TKs and farming experiences accumulated for years. Moreover, the use of early warning systems is not well developed in the area; farmers rely on TK and experience in farming to be the prime source of information on climate change and variability. On same study, farmers' perception of rainfall and temperature changes are compared against meteorological records, both sources confirmed similar results. In a study on drought monitoring using indigenous weather forecasting knowledge and meteorological records, consistency of drought events were observed in Guji, southern Ethiopia (Guye et al., 2022).

Participants of the FGD revealed that responsible bodies should work on proper documentation of TK of farmers on weather and climate issues. TK should be transferred to the coming generation as it helps to manage different environmental challenges facing their surroundings. Awareness creation for agriculture extension workers, youths and local development planners on the integration and link of TK with formal scientific knowledge, mainstreaming TK in the formal educational curriculum, and proper identification of TK scholars and work with them to address the various environmental and development challenges is frequently reported by FGD participants. Participants of KIIs also underlined that TK should be supported by science to increase its reliability and acceptance by the younger community. Moreover, the local government should support elders with TK to deliver better to the community. As frequently mentioned during discussions, farmers underlined documentation of TK and integration with scientific knowledge to help them to forecast and adjust to adverse weather conditions.

Makwara (2020) and Radeny et al. (2019) reported lack of systematic documentation and pass away of elderly and knowledgeable persons as challenges for the deterioration of TK accumulated for many years in different parts of the world. A recent study by Guye et al. (2022) on indigenous weather forecast among Guji pastoralists in southern Ethiopia, reported disappearance of indigenous weather forecasting knowledge due to growing generation gap religious teachings, technology, lack of clear transferring knowledge, poor documentation and education.

With all the challenges, traditional knowledge can benefit the SDGs as most goals are directly linked with it. Traditional knowledge is useful for sustainable land resource management, climate change adaptation, disaster risk reduction strategies and to address global challenges (Magni, 2017). Protection and inclusion of traditional knowledge in decision making processes is useful to manage weather and climate related risks. A study in the Bolivian Altiplano showed that traditional meteorological forecast knowledge has potential for improving agrometeorological forecasts and managing weather-related risks (Gilles et al., 2023).

4. Conclusions

This study explored TK of farmers on climate change and weather forecast in Menze Gera Midir district of Ethiopia. Combinations of metrological, biological, astrological, and other indicators are used to forecast forthcoming weather conditions. The use of TK reduced over time due to different challenge. This indicates the need for identification and documentation of TK, and integration in the formal educational system. While our study provides a useful description of farmers' TK of climate change and weather forecast, it does not

provide a meteorological explanation for the observed changes.

Author statement

AA has made substantial contributions in conception of the research and drafted the manuscript. AB and GA contributed in acquisition and analysis of data, and interpreted the results. AD reviewed and edited the manuscript.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

Data will be made available on request.

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Appendices.

Appendix 1Sample interview guide questions

- Do you think that climate is changing?
- What are the indicators that you confirmed climate is changing?
- How do you know the climate is changing?
- What are the sources of information about the changing climate?
- From whom/where you usually hear about the near future's weather condition? Is it from mass media; or listening from other farmers; or training and workshops; or agricultural extension workers; or your own experience; or a combination of two or more of these; or any other source?
- If your source of information includes own experience, how do you forecast the near futures weather condition? What indicators you use for forecasting the weather? Are there any religious and cultural or natural or any other ways of forecasting what will be the coming weather condition and related aspects?
- What biological, astrological, metrological and other indicators you use for forecasting the future weather conditions?
- What challenges do you are there any challenge you perceive or witness as barrier for traditional knowledge of weather forecasting?

Appendix 2. Sample focus group discussion questions

- Can you please explain the present weather (environmental) condition of your locality?
- Is there tendency of climate change in your environment?
- What is your observation on the climatic (temperature and rainfall) condition in your area in recent years?
- What metrological, ecological, or lifestyle changes you observed in recent years?
- Are the temperature and rainfall distribution and intensity changing? If yes how?
- What are the main climatic shocks (hazards) affecting livelihood in your locality? In the order of priority.
- In your community, how people know about whether there is a climate change or not?
- What is/are the source/s of information for the community in your area regarding weather and climate conditions?
- How do you know/forecast the coming weather conditions? Is it through modern weather forecasting? or Tradition knowledge through long years of experience? or using both?
- If you are using your own experiences in forecasting the coming weather conditions, what are these local knowledge experiences? What metrological, astrological, biological or other types of weather forecasting methods you use?
- While using these forecasting mechanisms, what challenges you face it might be from the scientific community or others?

Appendix 3
Reliability of the traditional indicators

List of indicators	Rank *	References (cases from other countries)

(*continued on next page*)

Appendix 3 (*continued*)

List of indicators	Rank *	References (cases from other countries)
Direction and strength of winds	1	Roncoli et al. (2002); Kasali (2011); Iticha and Husen (2019); Radeny et al. (2019)
Temperature change	2	Orlove et al. (2010); Chengula and Nyambo (2016); Radeny et al. (2019); Makwara (2020)
Color and movements of clouds	3	Orlove et al. (2010); Chengula and Nyambo (2016; Iticha and Husen (2019); Acharya and Prakash (2019; Tume et al. (2019); Gbangou et al. (2021); Nkuba et al. (2020)
Sound of hyena	4	We could not find a similar case in other countries
Behaviour of cattle	5	Mahoo et al. (2015); Tanyanyiwa (2018); Radeny et al. (2019)
Behaviour of donkeys	6	Radeny et al. (2019)
Behaviour of ants	7	Roncoli et al. (2001) Kalanda-Joshua et al. (2011); Iticha and Husen (2019); Radeny et al. (2019); Nkuba et al. (2020)
Behaviour of birds	8	Green et al. (2010); Leonard et al. (2013); Ifejika Speranza et al. (2010); Chang'a et al. (2010); Chengula & Nyambo (2016); Kolawole et al. (2014); Orlove et al. (2010); Roncoli et al. (2002); Kalanda-Joshua et al. (2011); Nkuba et al. (2020); Chengula and Nyambo (2016); Mapara (2009); Radeny et al. (2019)
Behaviour of chicken	9	Muguti and Maposa (2012)
Raphael market conditions	10	We could not find a similar case in other countries
Observation at the coffee ceremony	11	We could not find a similar case in other countries
Changes observed in a human body condition	12	Mahoo et al. (2015); Radeny et al. (2019)
Behaviour of bees	13	Chang'a et al. (2010); Galacgac and Balisacan (2009); Ifejika Speranza et al. (2010)
Behaviour of Butterflies	14	Wilken (1982); Nkuba et al. (2020)
Behaviour of plants	15	Ifejika Speranza et al. (2010); Ayal et al., 2015); Camacho -Villa et al. (2021)
Swirl winds	16	Radeny et al. (2019); Nkuba et al. (2020)

* = ranks are given for indicators as preferred by farmers based on perceived reliability and response of farmers.

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